



Dashboard > Courses > DIP321(English)(1),25/26-P > Topic 16 - Probabilistic Analysis and Randomized Algorithms > QUIZ - Topic 16 - Randomized Algorithms

Algoritmi un programmēšanas metodes(English)(1),25/26-P QUIZ - Topic 16 - Randomized Algorithms

Started on Thursday, 28 May 2026, 11:45 PM

State Finished

Completed on Thursday, 28 May 2026, 11:51 PM

Time taken 5 mins 53 secs

Marks 5.00/5.00

Grade **10.00** out of 10.00 (**100%**)

Question 1

Complete

Mark 1.00 out of 1.00

How can we use a randomized algorithm to help solve the hire assistant problem?

- ☐ a. By hiring the first candidate that meets a certain threshold of qualifications.
- ☐ b. By firing the current assistant and hiring a new one at random intervals.
- ☒ c. By selecting candidates to interview in a random order, rather than following the order given by the employment agency.
- ☐ d. By generating a random permutation of the candidates and selecting the best one.
- ☐ e. By interviewing all candidates and randomly selecting the best one.

Question 2

Complete

Mark 1.00 out of 1.00

What is the main idea behind probabilistic analysis?

- ☒ a. To use probability theory to model the behavior of the algorithm over a distribution of possible inputs.
- ☐ b. To analyze the worst-case performance of the algorithm.
- ☐ c. To perform experiments on the algorithm and measure its performance in various scenarios.
- ☐ d. To use random number generator to generate the answer for you.
- ☐ e. To analyze the average performance of the algorithm over all possible inputs.

Question 3

Complete

Mark 1.00 out of 1.00

What is a randomized algorithm?

- ☒ a. An algorithm whose behavior is determined by values produced by a random-number generator.
- ☐ b. An algorithm that uses random sampling to estimate solutions to problems.
- ☐ c. An algorithm that is guaranteed to produce the correct solution to a problem.
- ☐ d. An algorithm that is completely random.
- ☐ e. An algorithm that is deterministic and always produces the same output for a given input.

Question 4

Complete

Mark 1.00 out of 1.00

How does the Monte Carlo method differ from deterministic algorithms?

- ☐ a. Monte Carlo method only works in casinos.
- ☐ b. The Monte Carlo method is faster than deterministic algorithms and can solve problems of larger size.
- ☒ c. The Monte Carlo method uses randomness as a tool for solving problems, while deterministic algorithms always produce the same output for a given input.
- ☐ d. The Monte Carlo method can only be used for problems with a well-defined input distribution, while deterministic algorithms can handle any type of input.
- ☐ e. The Monte Carlo method is guaranteed to produce the correct solution to a problem, while deterministic algorithms may produce incorrect results in certain cases.

Question 5

Complete

Mark 1.00 out of 1.00

How is the Monte Carlo method used to estimate the value of pi?

- ☐ a. By generating a random permutation of the digits of pi and calculating the resulting value.
- ☐ b. By simulating creation of circles by random number of rectangles.
- ☒ c. By simulating a large number of random points in a square and calculating the proportion of those points that lie inside a quarter-circle inscribed in the square.
- ☐ d. By generating a random sequence of digits and counting the number of occurrences of the digit 3.
- ☐ e. By counting the number of primes in a randomly generated sequence of numbers.

[◀ Hiring Problem Presentation from Stanford](#)

Jump to...

[Survey - Lecture Speed ▶](#)